Forecasting with stationary models III

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Plan for this slide deck

1. Sources of forecast errors
2. Forecast evaluation
3. Forecast intervals

Feasible AR forecast

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t, \quad E[u_t \mid \{y_s\}_{s<t}] = 0$$

- We've discussed the optimal one-step-ahead forecast:

$$y_{T+1|T} = c + a_1 y_T + \cdots + a_p y_{T-p+1}.$$

- In practice, we don't know the parameters, so we substitute the OLS estimates. This yields a feasible forecast:

$$\hat{y}_{T+1|T} \equiv \hat{c} + \hat{a}_1 y_T + \cdots + \hat{a}_p y_{T-p+1}.$$

Forecast error

- Our forecast is never going to be exactly right.
- According to the model, the **forecast error** equals

$$y_{T+1} - \hat{y}_{T+1|T} = (c - \hat{c}) + (a_1 - \hat{a}_1) y_T + \cdots + (a_p - \hat{a}_p) y_{T-p+1} + u_{T+1}.$$

- The last term is the shock at time $T+1$. The uncertainty about this term never goes away, even if T is large.
- The other terms are due to estimation errors in the parameters. These errors vanish as the sample size T gets large.
- Of course, another source of error is model misspecification: it could be wrong to assume that y_{T+1} satisfies an AR(p) equation.
  - p too small, non-linearities, structural breaks...

Forecast quality

- A common way to measure the quality of a forecast $\hat{y}_{T+1|T}$ is the **mean-squared forecast error** (MSFE):

$$MSFE \equiv E[(y_{T+1} - \hat{y}_{T+1|T})^2].$$

- If we take the square root, we get the **root MSFE** (RMSFE):

$$RMSFE \equiv \sqrt{E[(y_{T+1} - \hat{y}_{T+1|T})^2]}.$$

- Of course, we could define similar concepts for multi-step forecasts:

$$MSFE(h) \equiv E[(y_{T+h} - \hat{y}_{T+h|T})^2].$$

Forecast evaluation

- How do we estimate the MSFE in the data?
- This is typically done through **recursive pseudo-out-of-sample forecasting**:

  1. First estimate the model using data only up to some time $\underline{t}$ in the past (e.g., 1947Q1–1969Q4). Then forecast the $\underline{t}+1$ observation (1970Q1). Compute the forecast error $\hat{e}_{\underline{t}+1|\underline{t}}$.
  2. Now *re-estimate* the model using data up to time $\underline{t}+1$ (1970Q1). Forecast the $\underline{t}+2$ observation (1970Q2) and compute forecast error $\hat{e}_{\underline{t}+2|\underline{t}+1}$.
  3. Etc. Stop at time T when we run out of data.
  4. We now have $T-\underline{t}$ forecast errors through time. Estimate the MSFE as

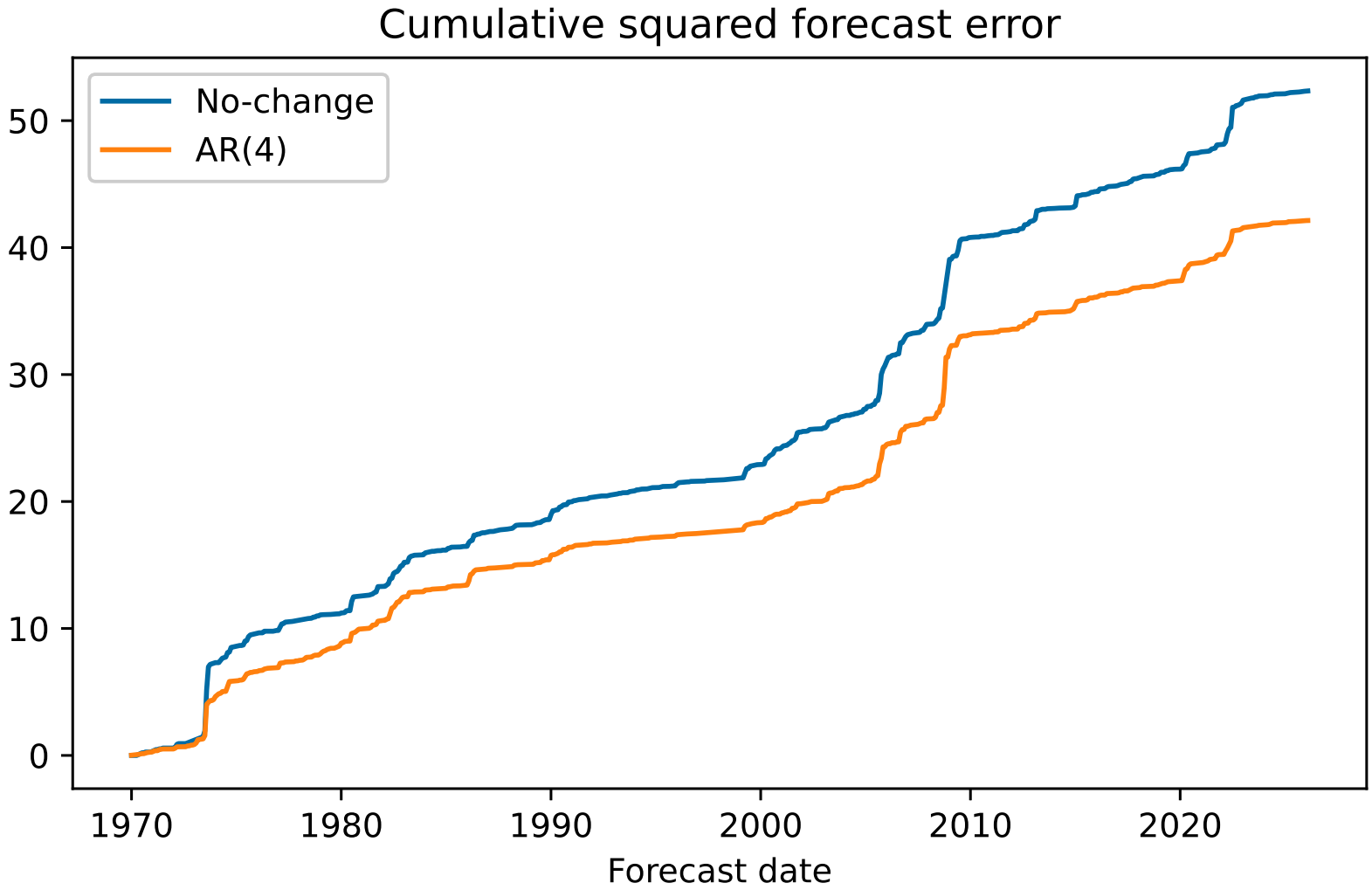
$$\widehat{MSFE} \equiv \frac{1}{T-\underline{t}} \sum_{t=\underline{t}}^{T-1} \hat{e}^2_{t+1|t}.$$

Comparing forecast methods

- The procedure on the previous slide uses real data, so does not assume that any particular model is the correct one that generated the data.
- Thus, if we are wondering which of several forecast methods to use, we can estimate their respective MSFEs as on the previous slide. Then pick the method with the lowest MSFE.

Comparing forecast methods: example

- Example: recursive evaluation of CPI inflation forecasts.
- Two competing methods: AR(4) vs. no-change forecast ($y_{t+1|t} = y_t$).
- We start the evaluation in January 1970, using data back to 1947 to estimate the AR model.
- Estimated RMSFE for AR(4): 0.250
- Estimated RMSFE for no-change forecast: 0.279
- For comparison, the sample std. dev. of month-over-month inflation is 0.340.


Cumulative squared forecast error
No-change
AR(4)
50
40
30
20
10
0
1970
1980
1990
2000
2010
2020
Forecast date

Extensions

- Alternative to MSFE: **mean absolute forecast error**

$$MAFE \equiv E[|y_{T+1} - \hat{y}_{T+1|T}|].$$

  Less sensitive to a few large forecast errors.

- The recursive pseudo-out-of-sample approach can easily be applied to multi-step forecasts, too.

- Both MSFE and MAFE give equal weight to positive and negative forecast errors. In some settings we may wish to weight over-predictions and under-predictions differently.

- If we're worried about structural breaks, we can estimate the model on a fixed sample size T_{window} at all time points, e.g., 20 years. This is called **rolling window** pseudo-out-of-sample evaluation.

Forecast interval

- So far we've produced **point forecasts**: a single "best guess" for what the future value of the time series will be.
- But these point forecasts will be associated with substantial uncertainty. The RMSFE is one measure of this uncertainty.
- In analogy with confidence intervals for parameters, we would like a **forecast interval** $\hat{\mathcal{I}}_{T+1|T}$ with the property

$$P(y_{T+1} \in \hat{\mathcal{I}}_{T+1|T}) \approx 1 - \alpha,$$

for some pre-specified coverage probability $1 - \alpha$.

Distribution of forecast error

- Let's go back to our model-implied forecast error

$$y_{T+1} - \hat{y}_{T+1|T} = (c - \hat{c}) + (a_1 - \hat{a}_1) y_T + \cdots + (a_p - \hat{a}_p) y_{T-p+1} + u_{T+1}.$$

- Now make the (strong) assumption that the shocks are i.i.d. normally distributed: $u_t \overset{iid}{\sim} N(0, \sigma_u^2)$.
- Recall that when T is large, parameter estimation errors are negligible relative to $\sigma_u^2 \equiv \text{Var}(u_{T+1})$.
  - We will later learn how to also take parameter uncertainty into account using Bayesian methods.
- Hence, as $T \to \infty$,

$$y_{T+1} - \hat{y}_{T+1|T} \overset{approx}{\sim} N(0, \underbrace{MSFE}_{\approx \sigma_u^2}).$$

Forecast interval: construction

- Given an estimate $\widehat{RMSFE}$ of $\sqrt{MSFE}$, we have

$$\frac{y_{T+1} - \hat{y}_{T+1|T}}{\widehat{RMSFE}} \overset{approx}{\sim} N(0,1).$$

- Let $Q_{1-\alpha/2}$ be the $1-\alpha/2$ quantile of the standard normal distribution (≈ 1.96 for $\alpha = 0.05$).

- Then

$$P\left(y_{T+1} \in \hat{\mathcal{I}}_{T+1|T}\right) \approx 1-\alpha, \quad \text{where}$$

$$\boxed{\hat{\mathcal{I}}_{T+1|T} \equiv \left[\hat{y}_{T+1|T} - Q_{1-\alpha/2}\widehat{RMSFE}, \hat{y}_{T+1|T} + Q_{1-\alpha/2}\widehat{RMSFE}\right].}$$

- Note: we're relying heavily on the assumption that the shock u_{T+1} is normally distributed. This is *not* a consequence of a CLT.

Forecast interval: estimating the RMSFE

- Two ways of obtaining an estimate $\widehat{RMSFE}$:
  1. Pseudo-out-of-sample forecasting.
  2. Model-based: $\widehat{RMSFE} = \sqrt{\hat{\sigma}_u^2}$ from the AR regression output.

Forecast interval: evaluation

- How do we evaluate the quality of a forecast interval?
- Using the pseudo-out-of-sample procedure, we can compute the fraction of the forecast intervals $\hat{\mathcal{I}}_{t+1|t}$ (constructed at each point in time) that actually cover their respective realized values y_{t+1}. This empirical coverage rate should be close to $1 - \alpha$.
  - Can use this to evaluate the assumption of normal shocks.
- Subject to having accurate coverage, we prefer forecast intervals that tend to be narrow. Can compute the average length of the intervals in the pseudo-out-of-sample evaluation.

2026:Q1 | 2025:Q4 | 2025:Q3 | 2025:Q2

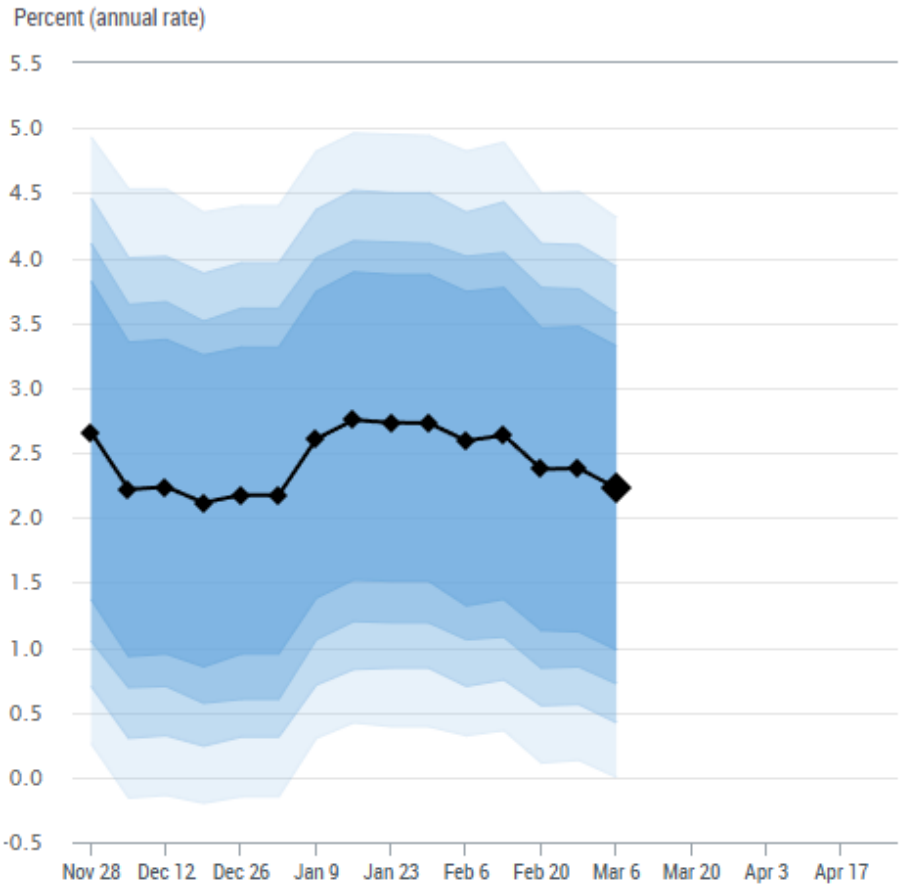
Latest Release: Mar 6, 2026

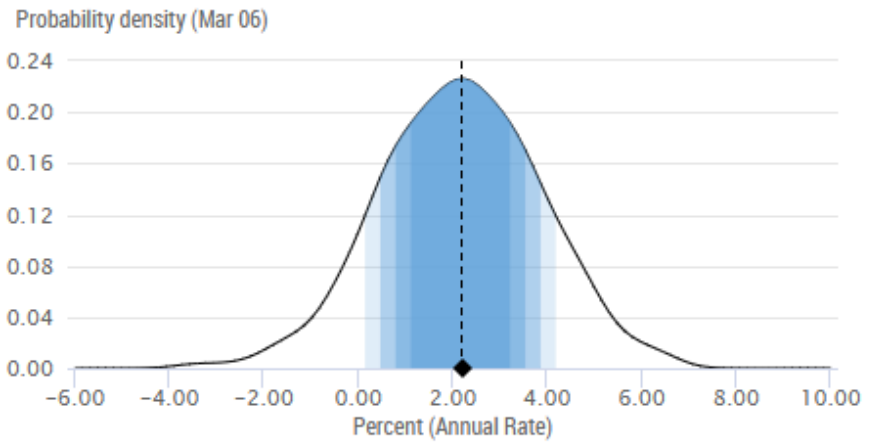
LAYOUT

New York Fed Staff Nowcast

◆ The New York Fed Staff Nowcast ■ 50% Probability ■ 60% Probability ■ 70% Probability ■ 80% Probability

□ Latest GDP Estimate ○ Advance GDP Estimate





Probability intervals

Release Date	Series	Value
Mar 6	◆ Nowcast	2.23
	■ 50% Probability	0.98, 3.32
	■ 60% Probability	0.72, 3.57
	■ 70% Probability	0.42, 3.93
	■ 80% Probability	0.00, 4.31
Feb 27	◆ Nowcast	2.38

Direct vs. iterated multi-step forecasts

- Recall: h-step forecast in AR(1) model:

$$\hat{y}_{T+h|T} = \hat{\rho}^h y_T + \frac{1-\hat{\rho}^h}{1-\hat{\rho}}\hat{c}.$$

  This is an **iterated** multi-step forecast, as it's obtained by iterating on the recursive AR model.

- Alternative approach: **direct** multi-step forecast regression

$$y_{t+h} = \hat{\beta}_0 + \hat{\beta}_1 y_t + \hat{v}_{t+h},$$

  yielding h-step forecast $\hat{y}^{\text{direct}}_{T+h|T} \equiv \hat{\beta}_0 + \hat{\beta}_1 y_T$.

- Direct forecasts are less subject to model misspecification bias, as they do not impose a recursive functional form.

- But their variance is higher. Typically $MSFE_{\text{direct}} > MSFE_{\text{iterated}}$.

What did we learn?

- Sources of forecast errors:
  1. Parameter uncertainty
  2. Unforecastable shock
  3. Model misspecification
- Forecast evaluation
  - (R)MSFE
  - Recursive pseudo-out-of-sample forecasts
- Forecast interval, assuming normally distributed shocks
- Direct vs. iterated multi-step forecasts

Looking ahead

- Allowing for richer time series dynamics
  - Moving averages
  - ARMA models
- Lag polynomials to facilitate analysis of rich time series models
- When is the AR(p) model stationary?